



University of Asia Pacific

Admit Card

Final-Term Examination of Fall, 2024

Financial Clearance

PAID

Registration No : 22201243

Student Name : H. M. Tahsin Sheikh

Program : Bachelor of Science in Computer Science and Engineering



SI.NO.	COURSE CODE	COURSE TITLE	CR.HR.	EXAM. SCHEDULE
1	CSE 301	Object Oriented Programming II and Web Programming	3.00	
2	CSE 302	Object Oriented Programming II and Web Programming Lab	1.50	
3	CSE 303	Computer Architecture and Organization	3.00	
4	CSE 305	System Analysis and Design	3.00	
5	CSE 307	Project Strategy and Management	0.00	
6	BUS 301	Business and Entrepreneurship	3.00	
7	ENG 300	Technical Writing and Presentation	1.50	
8	BIE 301	Bioinformatics Engineering	3.00	

Total Credit: 18.00

1. Examinees are not allowed to enter the examination hall after 30 minutes of commencement of examination for mid semester examinations and 60 minutes for semester final examinations.
2. No examinees shall be allowed to submit their answer scripts before 50% of the allocated time of examination has elapsed.
3. No examinees would be allowed to go to washroom within the first 60 minutes of final examinations.
4. No student will be allowed to carry any books, bags, extra paper or cellular phone or objectionable items/incriminating paper in the examination hall. Violators will be subjects to disciplinary action.

This is a system generated Admit Card. No signature is required.

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Fall 24
Program: B.Sc. in CSE
3rd year 1st Semester

Course Title: System Analysis and Design
Time: 3:00 hours

Credit Hour: 3.00

Course Code: CSE 305
Full Marks: 50

There are Five* Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. Differentiate between Agile System Development Life Cycle (SDLC) and the traditional SDLC. [10]

QUESTION 2 [10 MARKS]

- a. The optimistic time, pessimistic time and most likely time of a project are tabulated below. Calculate the probability that the project can be completed in 14 days. [10]

Activity	Dependency	Optimistic Time	Pessimistic Time	Most Likely Time p
A	---	2	4	3
B	A	3	6	4
C	A	4	7	5
D	B	1	3	2
E	C	2	6	3
F	D, E	1	4	2

QUESTION 3 [10 MARKS]

The initial investment for a project XYZ is Taka 50,000/=. It expects to provide revenues of Taka 13,000/=: Taka 26,000/= and Taka 23,000/= for the first, second and third years, respectively. The cost of capital is 13%. Analyze mathematically whether the investment is cost-beneficial or not. [10]

QUESTION 4 [10 MARKS]

- a. Illustrate the feasibility factors presented in the planning phase and the testing stages presented in the implementation phase of Prof. Hoffer's system development life cycle. [5]
- b. Explain how the requirements can be determined in order to develop a system according to the concept presented by Prof. Kendall. [5]

QUESTION 5 [10 MARKS]

- a. Design an online students' registration system using a sequence diagram. [10]
- OR
- b. Design an online shopping system using a class diagram. [10]

Start



Handwritten signature or initials.

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08691	.08534	.08379	.08226
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.0	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.1	.46017	.45620	.45224	.44828	.44433	.44038	.43644	.43251	.42858	.42465
-0.0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
2.0	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
3.0	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900
3.1	.99903	.99906	.99910	.99913	.99916	.99918	.99921	.99924	.99926	.99929
3.2	.99931	.99934	.99936	.99938	.99940	.99942	.99944	.99946	.99948	.99950
3.3	.99952	.99953	.99955	.99957	.99958	.99960	.99961	.99962	.99964	.99965
3.4	.99966	.99968	.99969	.99970	.99971	.99972	.99973	.99974	.99975	.99976
3.5	.99977	.99978	.99978	.99979	.99980	.99981	.99981	.99982	.99983	.99983
3.6	.99984	.99985	.99985	.99986	.99986	.99987	.99987	.99988	.99988	.99989
3.7	.99989	.99990	.99990	.99990	.99991	.99991	.99992	.99992	.99992	.99992
3.8	.99993	.99993	.99993	.99994	.99994	.99994	.99994	.99995	.99995	.99995
3.9	.99995	.99995	.99996	.99996	.99996	.99996	.99996	.99996	.99997	.99997

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Fall 24
Program: B.Sc. in CSE
3rd year 1st Semester

Course Title: Bioinformatics Engineering

Time: 3:00 hours

Credit Hour: 3.00

Course Code: BIE 301

Full Marks: 50

There are Five Questions. Answer all of them. Q3 has 2 options. Part marks are shown in the margins

QUESTION 1 [10 MARKS]

- a. Differentiate between the small-parsimony and large-parsimony problem. [2]
- b. Construct the De Bruijn graph from the following sequence using 3-mers. [5]
ATGCATTTGCATGCG
- c. Describe the flow of information from DNA to Protein using proper figure. [3]

QUESTION 2 [10 MARKS]

- a. Analyze the secondary structure of the RNA having the following sequence [7]
AGGGGUGACACUUACCCAAA
- b. Demonstrate the steps of the Divide and Conquer Approach to LCS. Explain how the alignment score can be computed with linear memory in this process. [3]

QUESTION 3 [10 MARKS]

- a. Apply the UPGMA algorithm to reconstruct a hierarchical cluster from the following distance matrix calculated upon human, seal, chimpanzee, panda, raccoon, and whale. [7]

$$D = \begin{vmatrix} 0 & 9 & 2 & 4 & 9 & 10 \\ 9 & 0 & 9 & 6 & 2 & 10 \\ 2 & 9 & 0 & 5 & 9 & 10 \\ 4 & 6 & 5 & 0 & 6 & 10 \\ 9 & 2 & 9 & 6 & 0 & 10 \\ 10 & 10 & 10 & 10 & 10 & 0 \end{vmatrix}$$

- b. Explain the meaning of the leaves, internal nodes, and root in an evolutionary tree. [3]

OR

- Ⓐ Apply the limb length calculation to reconstruct a phylogenetic tree from the following distance matrix calculated upon human, seal, chimpanzee, and panda. [7]

$$D = \begin{array}{c|cccc} & h & s & c & p \\ \hline h & 0 & 21 & 13 & 22 \\ s & 21 & 0 & 12 & 13 \\ c & 13 & 12 & 0 & 13 \\ p & 22 & 13 & 13 & 0 \end{array}$$

- Ⓑ Define the four-point condition with an example. [3]

QUESTION 4 [10 MARKS]

- Ⓐ Consider the following distance matrix on four taxa. First, determine if this is additive using the four-point condition; then draw the corresponding unrooted tree with branch lengths. You can use any of your preferred algorithms. [7]

$$D = \begin{array}{c|cccc} & 0 & 4 & 8 & 5 \\ \hline & 4 & 0 & 6 & 7 \\ & 8 & 6 & 0 & 11 \\ & 5 & 7 & 11 & 0 \end{array}$$

- Ⓑ Perform rooting on this unrooted tree and express it as Newick notation with branch lengths. [3]

QUESTION 5 [10 MARKS]

- a. Explain the difference between Hamming Distance and Edit Distance. [2]

- Ⓑ Apply Greedy Sorting by Reversal algorithm for finding genome rearrangement on these syntenic blocks. Show every step. [5]

- 4 + 7 + 5 - 1 - 3 + 6 - 2 + 9 - 10 - 8

- c. Describe the process of mutation and its effects on the organism. [3]

+1 5 7 4 -3 +6 -2 8 10 -9
1 5 7 3 -4 +6 -2 8 9 -10
1 2

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Fall 24
Program: B.Sc. in CSE
3rd year 1st Semester

Course Title: Computer Architecture & Organization

Time: 3:00 hours

Credit Hour: 3.00

Course Code: CSE 303

Full Marks: 50

There are Five Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. Demonstrate and implement the single cycle Datapath for the following instructions and write down the procedure of Datapath for following MIPS instructions: (consider all the functional units of Datapath according to MIPS architecture) [10]

i) lw \$t0, 32(\$s0);

ii) sub \$s0, \$s1, \$s0;

iii) sw \$t1, 48(\$s1);

QUESTION 2 [10 MARKS]

- a. Consider a non-pipelined machine with 5 execution stages of lengths 2 ns, 5 ns, 4 ns, 6 ns, and 3 ns. [5]

i) Detect the instruction latency on this machine and how much time does it take to execute 1000 instructions

ii) Suppose a pipeline is introduced here. Now calculate the instruction latency on the pipelined machine. How much time does it take to execute 1000 instructions now? Consider 1ns slack time in the pipeline.

iii) Calculate the speed-up.

- b. This question considers the basic MIPS, 5-stage pipeline (IF, ID, EXE, MEM, WB). [5]

Assume that you have the following sequence of instructions:

- i) lw \$t0, 32(\$s0);
- ii) sub \$s0, \$s1, \$s0;
- iii) sw \$t1, 48(\$s1);

Illustrate the implementation through 5 stages and explain the implementation for both pipelined and non-pipelined design briefly. Mention if there is any pipelining hazard.

QUESTION 3 [10 MARKS]

- a. Illustrate the Booth's Algorithm step by step to multiply:
 $5 * -3$ Using a 4 bits multiplier. Consider -3 as the multiplier. [7]
- b. Diagram the algorithm flow graph of the 3rd version of the division algorithm. [3]

QUESTION 4 [10 MARKS]

- a. CPU requests the following Block addresses 8, 21, 9, 17, 20, 21, 16, 8, 22 and 25. There are 16 blocks in cache. Illustrate the memory mapping for the following cache configurations: [3+
7]
- I. Direct mapped
- II. 2-Way, 4-way, 8-way and 16-way set associative mapped.

OR

- a. I) Diagram the cache Hit and miss process for direct mapping function of MIPS architecture. [3+
7]
- II) Diagram and illustrate all the functional elements of MIPS processor(both data hold and data operate elements) .

QUESTION 5 [10 MARKS]

- a. Instruction and Opcode/Function: lw 100011, sw 101011, sub 100010, add 100000 [10]

Analyze the following high-level statement according to MIPS instruction format and write the MIPS Assembly code and machine Code.

$$A[101] = B - C[15] + D;$$

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Fall 24
Program: B.Sc. in CSE
3rd year 1st Semester

Course Title: Object Oriented Programming II: Visual and Web Design
Time: 3:00 hours

Credit Hour: 3.00

Course Code: CSE 301

Full Marks: 50

There are **Five*** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. Take a list of integers (both positive and negative) from the user and send this list to a function named **find_greater**. Inside the **find_greater** function, first ignore all the negative numbers and only work with the positive numbers from the list. Then, calculate the sum of all even positive numbers and the sum of all odd positive numbers. After calculating both sums, compare them. Return "Even sum is greater", "Odd sum is greater", or "Both sums are equal" with the sum, based on which is larger or if they are equal. [4]

Sample Input	Sample Output
2 4 -1 3	Even sum is greater which is: 6

- b. Write a Python program that opens a file named "data.txt", reads its content, and finds the first occurrence of a word (entered by the user). The program should then display the line number where the word first appears. If the word is not found, the program should display "Word not found in the file!". If the file is not found, display "File not found!". [6]

Sample File	Sample Input	Sample Output
This is a text file. Python is a powerful programming language. Many developers love Python. Python can be used for various applications.	Enter the word to search: python	The word 'python' first appears in line 2.

QUESTION 2 [10 MARKS]

You are developing a system called PassionTracker, which helps people track and manage their hobbies and passions (like painting, coding, writing, etc.). Every passion must follow a common structure, but each passion implements the details in its own way.

To achieve this, you decide to use abstract methods in an abstract class so that every subclass is forced to implement specific behaviors like `start()`, `practice()`, and `showProgress()`.

- a. Create an abstract class named `Passion` that contains three abstract methods: [4]

- `start()`
- `practice()`
- `showProgress()`

Then, create a subclass called `Painting` that inherits from `Passion` and provides actual implementations for each abstract method. In the `Painting` class, define each method to print a meaningful message (e.g., "Started painting", "Practicing brush strokes", "Showing painting progress").

- b. In your `Painting` class, add a private data member named `__progress` (an integer). [6]

Write methods to safely access and update this private variable:

- A public method `updateProgress(hours)` that increases `__progress` by the number of hours spent practicing.
- A public method `getProgress()` that returns the current progress.

Also, explain why it is important to make `__progress` private instead of public in this OOP design.

OR

- a. How many types of access modifiers are there in Python? List them and briefly describe each one with code. [4]

- b. Demonstrate with code how protected, and private variables can be accessed from outside a class in Python. Explain why access modifiers in Python are a matter of developer discipline rather than strict enforcement by the language. [6]

QUESTION 3 [10 MARKS]

- a. Enumerate and explain the different types of Cascading Style Sheets (CSS) with suitable examples for each type. [4]

- b. **Welcome to My Homepage** [6]

Product Table

Product	Price	Available
Notebook	\$10	Yes
	\$12 (with cover)	Yes
Pen	\$1	No

Features

- User-friendly interface
- Fast loading
- Responsive layout

Contact Us

Name:

Email:

Message:

Write an HTML code to implement the homepage as described above.

QUESTION 4 [10 MARKS]

An admin of an online bookstore wants to add a new book to the store's catalog by submitting a form that includes fields such as *Title*, *Author*, *Price*, and *Description*.

- a. Describe how you would implement this functionality in Django using the MVT (Model-View-Template) pattern. [4]

- b. Write sample Django code snippets to define the model for the book, create the HTML form, and handle the form submission in the view to save the data to the database. [6]

QUESTION 5 [10 MARKS]

- a. Explain with code, how to work with media files in Django. Include the necessary steps and code snippets to handle media file uploads, including configuration settings, models, and any other relevant setups. [4]

- File* b. You are building an e-commerce website where users can rate products, but each user can only rate a product once. When rating, the user should not select a product; instead, the system automatically associates the rating with the product the user is currently viewing. Define a Product model and a ProductRating model, ensuring that the ProductRating allows a user to rate a product only once. The rating should be automatically linked to the logged-in user and the specific product they are viewing. Create a ProductRatingForm in forms.py for submitting the rating without the user selecting a product, and set the current user as the customer of the ProductRating class. Implement view logic in views.py so that when a user is viewing a product, the system automatically associates the ProductRating with that specific product. If the user has already rated the product, they should be informed that they cannot rate it again. [6]

You need to define:

- **models.py** (Product, ProductRating)
- **forms.py** (ProductRatingForm)
- **views.py** (handle ProductRating function)
- **products.html** (Handle how to view all the products and a button to rate each one of them)

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Fall 24
Program: B.Sc. in CSE
3rd year 1st Semester

Course Title: Project Management & Strategy
Time: 3:00 hours Credit: 0.0

Course Code: CSE 307
Full Marks: 50

There are **Five*** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. Discuss the importance of a good communication system. [5]
- b. Identify some key practices to develop effective communication skills. [5]

QUESTION 2 [10 MARKS]

- a. Describe the necessity of decision trees in quantitative risk analysis. [5]
- b. Classify six major processes involved in risk management. [5]

QUESTION 3 [10 MARKS]

- a. TechNova Electronics is a mid-sized company that assembles smartphones using parts from multiple international suppliers. The company is facing increasing costs due to poor coordination with suppliers, delays in delivery of parts, and challenges in forecasting the right quantity of components needed for production. Some production lines have stopped due to missing components, while others have excess stock. The company wants to improve coordination and reduce inefficiencies across its supply and production process. [10]

Categorize the type of Management Information System TechNova should implement to improve coordination with suppliers and optimize inventory levels? Illustrate the system and how such a system could help.

OR

[10]

- a. RXTextiles Ltd. is a rapidly expanding textile company. Different departments like finance, HR, procurement, and inventory use separate, unconnected software systems. As a result, data is inconsistent and reporting is delayed. Managers often struggle to make timely decisions due to a lack of unified information. The company is looking for a system that can integrate its core business functions to support real-time planning and decision-making.

Categorize the type Management Information System should RXTextiles implement to unify its departments and improve overall business operations? Illustrate the system and how such a system could help.

QUESTION 4 [10 MARKS]

A software company is planning to develop two products: Project Alpha is estimated at 350 KLOC and is classified as Organic. Project Beta is estimated at 500 KLOC and is classified as Semidetached.

- a. Calculate the effort and development time for both projects using the Basic COCOMO model. [5]
- b. Evaluate which project is more efficient in terms of effort per KLOC and explain why. [5]

QUESTION 5 [10 MARKS]

Imagine you are managing a software project in a startup. Your team is highly skilled, but you have limited funding and weak brand recognition. There is rising demand for your app, but several large competitors are entering the same market.

- a. Perform a SWOT analysis based on this situation. [5]
- b. Illustrate two short term and two long term project strategies based on your analysis to improve project outcomes. [5]

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Fall 24
Program: B.Sc. in CSE
3rd year 1st Semester

Course Title: Business and Entrepreneurship
Time: 3:00 hours

Credit Hour: 3.00

Course Code: BUS 301
Full Marks: 50

Part A

Answer any of the two (02) questions.

1. SRF Company is considering three projects (A, B, and C) that are mutually inclusive.

Year	Project A BDT	Project B BDT	Project C BDT
0 (Initial Investment)	(250,000)	(300,000)	(300,000)
1	100,000	120,000	140,000
2	110,000	130,000	150,000
3	120,000	140,000	160,000
4	60,000	160,000	180,000

Conditions & Constraints:

- i. The total budget for selected projects must not exceed BDT 550,000.
 - ii. The company wants to select projects that provide the NPV.
 - iii. The discount rate is 10%.
- a. Calculate the NPV for each project individually. 10
- b. Determine which project/projects should be selected based on the budget constraint. If the company finds a way to increase its budget to BDT 600,000, would the decision change? Why or why not? 05

2. At the beginning of February 2024, TFA Company had the following account balances:

- Cash: BDT 60,000
- Accounts Receivable BDT 5,000
- Equipment: BDT 60,000
- Accounts Payable: BDT 15,000
- Bank Loan: BDT 30,000
- Owner's Equity: BDT 80,000

During the Month, the Company:

- Feb 01: Purchased equipment for BDT 20,000 in cash.
Feb 12: Purchased BDT 25,000 on credit from Mr. Rakiv (Supplier).
Feb 13: Sold goods cash for BDT 45,000.
Feb 14: Paid bank loan BDT 2,000.
Feb 13: Sold goods cash for BDT 50,000 out of which BDT 35,000 was received in cash and BDT 15,000 on account.
Feb 13: Paid transportation expense BDT 2,500.
Feb 20: Paid BDT 1,000 for advertising expenses.
Feb 28: Paid salaries to employees BDT 6,000.

a. Prepare the tabular analysis from the above transactions using the accounting equation (Assets = Liabilities + Owner's Equity). 10

b. Prepare the "Income Statement" and "Balance Sheet".

[3+2]

3.

SRF Company
Statement of Financial Position
As at 31 December 2023 and 2022

Assets	2023	2022
Current assets	BDT	BDT
Cash	420,700	449,400
Prepaid expenses	518,000	350,000
Accounts receivable	824,600	719,600
Inventory	882,000	808,500
Total current assets	2,645,300	2,327,500
Plant assets (net)	4,543,000	3,642,100
Total assets	<u>7,188,300</u>	<u>5,969,600</u>
Liabilities and Stockholders' Equity		
Current liabilities		
Accounts payable	1,120,000	1,015,000
Income taxes payable	304,500	294,000
Total current liabilities	1,424,500	1,311,800
Bonds payable	1,540,000	1,400,000
Total liabilities	2,964,500	2,711,800
Stockholders' equity		
Common stock (BDT 5 par)	2,030,000	2,100,000
Retained earnings	2,193,800	1,157,800
Total stockholders' equity	4,223,800	3,257,800
Total liabilities and stockholders' equity	<u>7,188,300</u>	<u>5,969,600</u>

SRF Company
Statement of profit or loss and other comprehensive income
For the year ended 31 December 2023 and 2022

Particulars	2023 (BDT)	2022 (BDT)
Revenue	630,000	525,000
Cost of the service	367,500	315,000
Gross Profit	262,500	210,000
Operating Expenses	122,500	105,000
Net Income	<u>140,000</u>	<u>105,000</u>

- a. Calculate each ratio mentioned below for both years and explain the result of each ratio: [5+5]

- i. The current ratio
- ii. The acid-test ratio
- iii. Debt to equity
- iv. Profit Margin Ratio
- v. Gross Profit Rate

- b. Mention the **benchmark** of the mentioned five ratios and which year the company performed better. 05

4. a. Define the break-even point. Why is it important for startup founders to understand this concept? 05

- b. A startup incurs monthly **fixed costs** of BDT 500,000. It sells its product at **BDT 1,000 per unit**, and the **variable cost per unit** is BDT 250. 10

The founder is considering two pricing strategies:

- **Option A:** Keep the price at BDT 1,000 per unit
- **Option B:** Increase the price to BDT 1,200 per unit to improve margin

Tasks:

1. Calculate the **break-even sales volume** under **Option A**.
2. Calculate the **break-even sales volume** under **Option B**.
3. Briefly explain which pricing strategy seems better from a break-even perspective, assuming the market can absorb the price increase.

Part B

Answer any of the two (02) questions.

Global Smartphone Market (2025)

The smartphone industry remains one of the most dynamic sectors in the world economy. In 2025, it is estimated to be worth around \$600 billion, with major players like Apple, Samsung, Xiaomi, Oppo, and Vivo controlling a significant share of the market. Despite its maturity, the market still sees around 4% annual growth.

Entering the market is not easy. New brands must overcome huge investments in research and development, build strong brand recognition, and deal with a complex web of supplier relationships. Intellectual property issues are also a major hurdle, with giants like Apple and Samsung holding key patents.

Suppliers of key components, particularly chips and displays, hold considerable importance. However, large smartphone brands often source from multiple suppliers to avoid dependency. For example, companies like Apple work with TSMC for chips and Samsung Display for screens but are constantly expanding their supplier base.

On the consumer side, buyers are well-informed and increasingly price-sensitive. Easy comparisons, online reviews, and seasonal discounts give customers a lot of power in choosing between brands. Mobile network carriers (like Verizon, AT&T, Vodafone) still play an important role in promoting specific models through bundled offers.

Substitute products like refurbished phones, feature phones, and even wearable devices that allow texting and calling have started eating into traditional smartphone sales, especially in price-sensitive markets like India and Africa. While they don't yet threaten high-end smartphone sales, they do affect volume growth at the lower end.

Competition within the industry is intense. Most brands release new flagship models every 12 months or faster. Margins are getting squeezed as brands try to offer better specs at lower prices. Marketing, software ecosystems (like iOS and Android customizations), and customer service are major areas of differentiation.

Environmental regulations are becoming more serious. Companies are being pushed to reduce electronic waste and incorporate recycled materials into production. Privacy concerns and cybersecurity are also shaping how brands develop and market their devices.

- | | | |
|----|--|----|
| 5. | Analyze the smartphone industry using Porter's Five Forces framework | 10 |
| 6. | a. Define the four major types of market segmentation. Provide one brief example for each based on a digital product or service. | 05 |
| | b. You are launching a mobile health tracking app for working professionals in Dhaka. | 05 |
| | • Identify which segmentation type you would prioritize. | |
| | • Based on that, create a brief customer persona including at least three characteristics (e.g., age, lifestyle, goals). | |

Explain why this persona is a good fit for your chosen segment.

Company Overview: EcoSip

EcoSip launched in 2015 in California with a mission to offer healthier beverage options while caring for the environment. Over a short span, it carved out a strong niche, with its colorful organic juices and sustainably packaged water now found in over 4,000 stores across the U.S. Customers are drawn to the brand's creative flavors like blueberry-ginger water and turmeric-lime juice, and its promise to use 100% biodegradable or recyclable packaging. EcoSip also stands out for building direct relationships with organic farms, ensuring quality and freshness in its products. Every year, it donates 2% of profits to environmental causes, which has built a loyal customer base that believes in its mission.

Still, EcoSip faces several challenges. Its production costs are high compared to larger beverage companies, and its premium pricing sometimes limits appeal beyond niche markets. Distribution is another bottleneck — without deals with giants like Walmart or Target, the brand relies heavily on smaller retailers. There's also pressure from limited marketing resources, making it harder to compete for visibility against bigger brands with massive advertising budgets.

The road ahead offers exciting possibilities. The health and wellness trend is showing no signs of slowing, and new international markets are opening up to organic products. EcoSip could partner with fitness centers, boutique hotels, or premium grocery chains to broaden its presence. The company's plan to launch a new sparkling water line, designed to appeal to a younger, trend-conscious audience, is another step toward growth.

However, risks are real. Major players like Coca-Cola and PepsiCo are rapidly expanding their organic and "natural" beverage lines, offering lower prices that could lure away EcoSip's customers. Economic downturns could also hit consumer willingness to pay premium prices. On top of that, raw material costs for organic fruits are volatile, threatening profitability if prices swing too much.

- a. Based on EcoSip's story, prepare a detailed **SWOT analysis** identifying at least three points under each category. 07
- b. **Discuss three types of Competitors.** 03
8. a. "There are three types of crises that a startup might face" Briefly explain how each can impact business operations. 06
- b. Describe any four practices that a founder can adopt to build a more resilient startup. 04
9. Explain in detail the four elements of the **marketing mix (4Ps)** with examples. 10